

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda · Remote Worldwide

PROFESSIONAL SUMMARY

Backend engineer with **6+ years of production Node.js**, combining hands-on **technical implementation ownership** with **team leadership**. Took responsibility for the end-to-end migration of 20+ applications to Firebase/GCP with zero downtime reporting directly to the CTO, designing the real-time Pub/Sub sync layer that kept both systems live through the cutover and cutting infrastructure spend by \$5,000/month. **Led a team of 4 developers** and **architected a microservices backend** on Apache Kafka, Docker and Kubernetes. Deep experience across **API design, database system design and launching applications from scratch**, with hands-on AWS (S3/EC2/EBS) and GCP. Six years delivering remotely for international clients in the US and Singapore. Based in Kampala (UTC+3) — full overlap with European teams. GitHub: github.com/zac-09

CORE COMPETENCIES

Node.js Backend Development

Technical Team Leadership

API Design & Integration

Database System Design

Microservices Architecture

AWS & GCP

Zero-Downtime Migrations

Mentoring & Developer Enablement

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- **Took responsibility for the technical implementation** of an end-to-end migration of 20+ applications from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Designed and built real-time two-way sync between MongoDB and Firestore using **Node.js** and Google Pub/Sub message processing, so both systems served live production traffic throughout the cutover.
- Owned the cloud cost side: saved the company **\$5,000/month** by migrating all services off **AWS**, and moved file storage from **Amazon S3** to GCS via a Python script.
- Built a 3D visualisation feature for customers using video overlay and the ffmpeg library, increasing buyer conversion.
- Implemented Express Shipping, generating an additional **\$40 revenue per order**.
- Migrated a WebFlow website from a Parse backend to Firebase.

Stack: Node.js, Firebase, Firestore, GCP Pub/Sub, AWS S3/EC2/EBS, MongoDB, Python, ffmpeg, React, JavaScript

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Hired via Upwork to execute a complex backend migration from Parse/MongoDB to Firebase Firestore and GCP.
- Built a migration script reading from MongoDB and writing to Firestore after complex processing logic — a cross-engine **database system redesign**.
- **Mentored and enabled the wider team**: wrote documentation for fellow engineers on the new Firebase SDKs (Node.js and Python) and trained the Ops team on the Firebase Dashboard and backup procedures.

Stack: Node.js, Firebase, Google Cloud Platform, AWS S3/EC2/EBS, MongoDB, JavaScript

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led optimisation of a PWA delivering real-time stock market prices from a Firebase backend.
- Built ad-hoc jobs to query over **2 million Firestore records**, keeping customer prices current from **Morningstar APIs**.
- Extended backend **APIs** using Firebase Cloud Functions and **Express.js** hosted on Heroku.
- Added responsive UI pages to the PWA using HTML, Tailwind, and React.

Stack: Node.js, Express.js, Firebase, React, Tailwind, HTML, CSS, JavaScript

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Full-stack delivery for a US-based **agency with demanding clients**, reporting directly to the CTO.
- Built backends for DeFi applications using Web3 and **Node.js**.
- Delivered React UIs from Figma designs under tight timelines.

Stack: Node.js, Firebase, React, Tailwind, HTML, CSS, JavaScript

Software Engineer · Uganda Remote

- **Led a team of 4 developers** building legal tech systems for NGOs in Uganda, delivered from scratch.
- **Architected a microservices backend** for FIDA Uganda's case management app using Apache Kafka, Docker, and Kubernetes.
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda.
- Developed a cross-platform mobile app (iOS & Android) for LASPNET using React Native and Expo, with Firebase push notifications.
- Built a USSD service for legal aid providers communicating with a Node.js backend via **gRPC**.

Stack: Node.js, Apache Kafka, Docker, Kubernetes, NATS Streaming, gRPC, React Native, Firebase, JavaScript

PROJECTS

Parse/MongoDB → Firebase Migration Programme Technical Ownership

Owned the technical implementation of a 20+ application migration to Firebase/GCP with no service interruption, reporting to the CTO. Ran both stacks in parallel behind a real-time two-way Pub/Sub sync layer and cut infrastructure spend by \$5,000/month.

Node.js · Firebase · Firestore · GCP Pub/Sub · MongoDB · AWS S3 · Python

FIDA Uganda Case Management Platform Architecture + Team Lead

Greenfield microservices platform for a legal-aid NGO. Architected the backend and led 4 developers through delivery, alongside a cross-platform React Native client and a gRPC-backed USSD service for feature-phone users.

Node.js · Apache Kafka · Docker · Kubernetes · gRPC · NATS Streaming · React Native

Real-Time Market Data Platform API + Data Design

Kept customer-facing stock prices current across 2M+ Firestore records by integrating third-party Morningstar APIs, with Express.js and Cloud Function endpoints behind a responsive React PWA.

Node.js · Express.js · Firebase Cloud Functions · Firestore · REST APIs · React

EDUCATION

BSc Computer Engineering — Makerere University

Aug 2018 – Sep 2022

Kampala, Uganda

SKILLS

Backend: Node.js, Express.js, REST APIs, gRPC, Apache Kafka, microservices architecture

Leadership: Team leadership, mentoring, architecture ownership, technical documentation, developer enablement

Data: MongoDB, Firestore, SQL, Google Pub/Sub, NATS Streaming, data modelling & migration

Cloud & DevOps: GCP, Firebase, AWS (S3/EC2/EBS), Docker, Kubernetes, Linux, Git **Languages:** JavaScript, Python, Go, Rust

Frontend: React, React Native (Expo), Tailwind, HTML, CSS **Spoken:** English (fluent), Kiswahili